

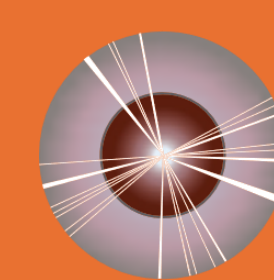
Integrating Triple-View Chemical Graph Aggregation with Deep Topological Embedding for ADME Property Prediction

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Introduction

Motivation

- Accurately representing chemical structures is critical for drug discovery, toxicity prediction, and the analysis of molecular property.
- Single molecular graph representation cannot fully model interactions between key substructures such as functional groups, scaffolds, and stereochemistry.
- Limitations in capturing substructural complexity require refined molecular representation frameworks.

Goal

- To propose a Triple-View graph by integrating substructure information from three distinct decomposition techniques: BRICS[1], and Junction Tree[2], Murcko scaffold[3].
- To effectively capture structural context within molecules and enhance expressive power across diverse prediction tasks.

Datasets

Pre-training

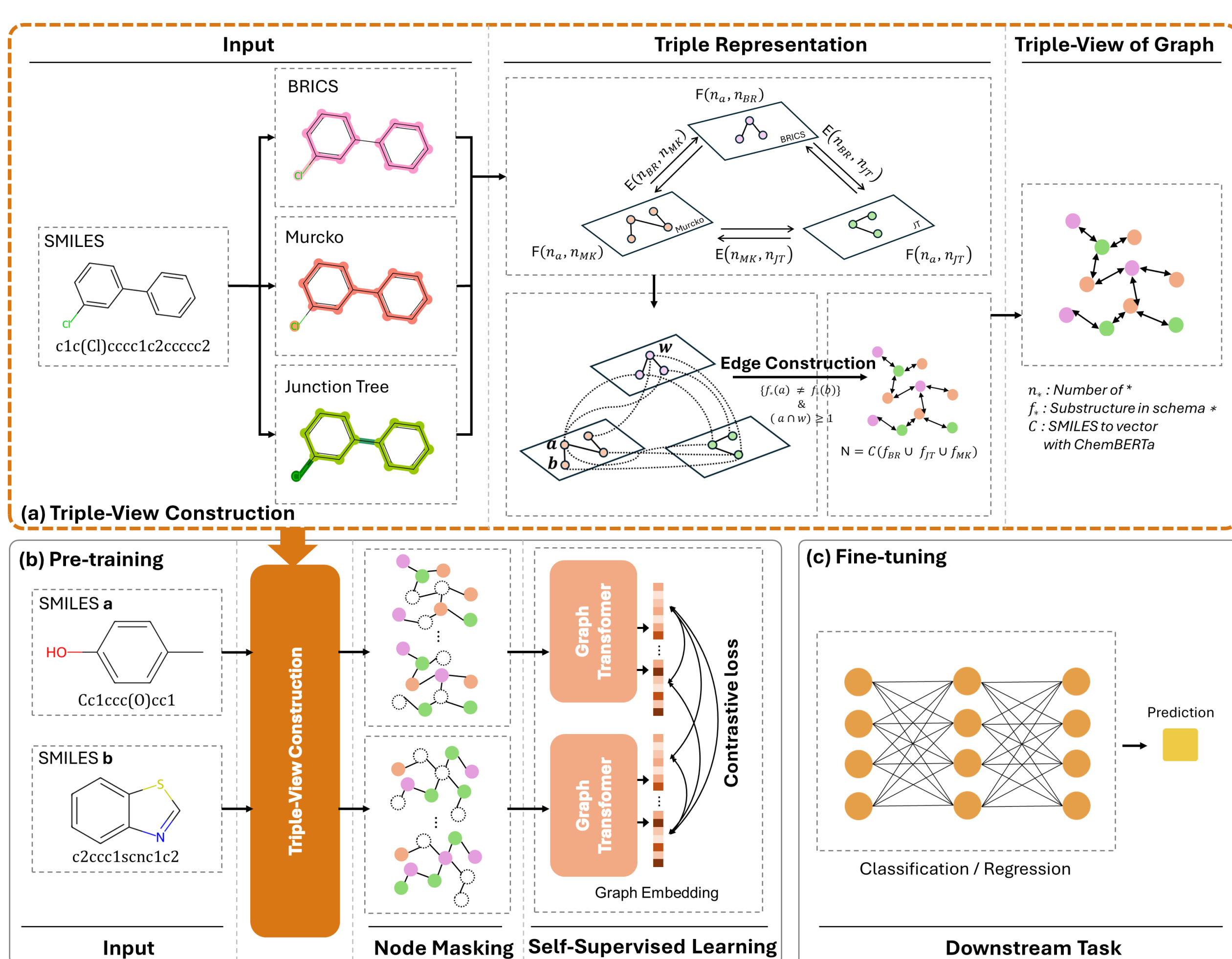
- Pretraining of the Triple-View graph model on **ZINC250k**[4].
- Diverse molecular set in ZINC250k covering broad chemical space for learning generalizable representations.

Fine-tuning

- Using 6 benchmark datasets from **MoleculeNet benchmark datasets**[5], covering both classification and regression tasks.
- Dataset split using the standard data splits (e.g., scaffold split, random split) defined by MoleculeNet into training, validation, and test sets.

Dataset	Task Type	# Tasks	# Compounds	Metric
BBBP	Classification	1	2,039	ROC-AUC
BACE	Classification	1	1,513	ROC-AUC
HIV	Classification	1	4,1127	ROC-AUC
Lipophilicity	Regression	1	4,200	RMSE
ESOL	Regression	1	1,128	RMSE
FreeSolv	Regression	1	642	RMSE

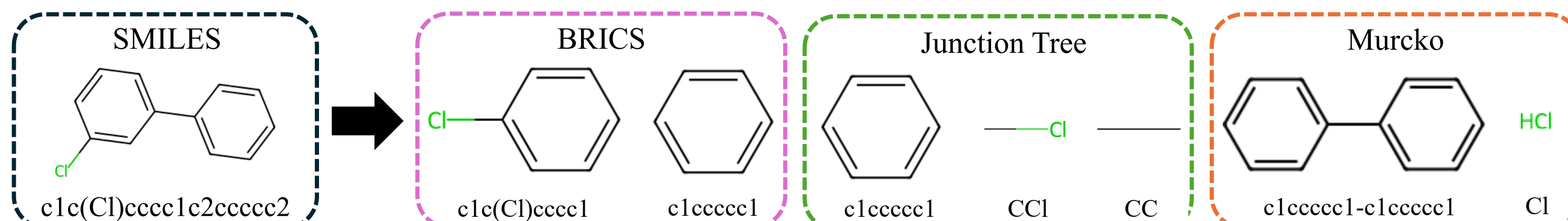
Method Overview: Triple-View Graph-based Molecular Representation Learning



(a) Triple-View Construction

Input

- Decomposition of each input SMILES using three distinct methods: **BRICS**, **Junction Tree**, **Murcko**.



- The extracted substructures serve as nodes in the Triple-View of graph.

Triple Representation

- Nodes are connected by edges weighted using the overlap weight matrix $W_{X,Y}[i,j] = \begin{cases} \frac{|F_{X,i} \cap F_{Y,j}|}{|F_{X,i}|} & i < j \\ \frac{|F_{X,i} \cap F_{Y,j}|}{|F_{Y,j}|} & i > j \\ 0 & i = j \end{cases}$

where F_X, F_Y are substructures from different decomposition methods and i, j denote specific substructure nodes in F_X and F_Y respectively.

- Using $W_{X,Y}$, the combination of substructure nodes and weighted edges creates three interaction graphs G capturing cross-method relationships: $G_{(BR, JT)}, G_{(JT, MK)}, G_{(BR, MK)}$
- These are merged into the Triple-View graph: $G_{triple} = G_{(BR, JT)} \cup G_{(JT, MK)} \cup G_{(BR, MK)}$

(b) Pre-training

Node Masking: Generation of two distinct views via random masking in the Triple-View graph.

Self-Supervised Learning: Encoding of molecular and contrastive learning of graph embeddings.

(c) Fine-tuning: Downstream task using pretrained embeddings as input.

Results

1. Benchmark Comparison

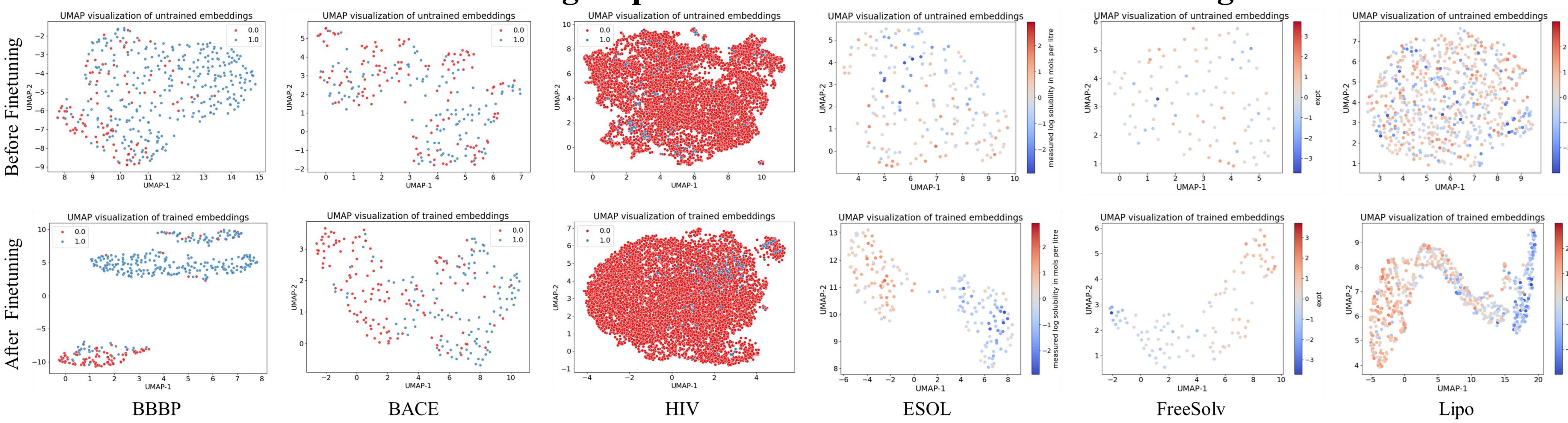
- Robustness across all benchmark datasets.
- SOTA performance on ESOL and FreeSolv datasets.
- Competitive results, closely matching or surpassing most baselines.

Method	Classification (ROC-AUC↑)			Regression (RMSE↓)		
	BBBP	HIV	BACE	ESOL	FreeSolv	Lipo
GCN	0.877±0.036	-	-	1.068±0.050	2.900±0.135	0.712±0.049
MPNN	0.913±0.041	-	-	1.167±0.430	2.185±0.952	0.672±0.051
GROVER	0.911±0.008	-	-	0.911±0.116	1.987±0.072	0.643±0.030
MoICLR	0.738±0.020	-	0.788±0.050	1.16±0.00	2.39±0.14	0.78±0.01
MvMRL	0.963±0.019	-	0.878±0.336	0.856±0.088	1.770±0.229	0.686±0.037
HIGNN	0.932±0.017	0.816±0.026	0.890±0.024	0.532±0.064	0.915±0.189	0.549±0.034
Group Graph	0.931±0.006	0.821±0.016	0.886±0.015	0.666±0.060	1.37±0.28	0.654±0.013
MGSSL(DFS)	0.705±0.011	0.795±0.011	0.797±0.008	-	-	-
MGSSL(BFS)	0.697±0.009	0.788±0.012	0.791±0.009	-	-	-
FMGCL	0.951±0.023	-	0.866±0.047	0.751±0.092	1.976±0.526	0.618±0.056
MoIFCL	0.981±0.002	0.855±0.020	0.933±0.010	0.638±0.024	1.045±0.160	0.601±0.034
Ours	0.929±0.014	0.738±0.015	0.597±0.011	0.521±0.014	0.582±0.036	0.785±0.026

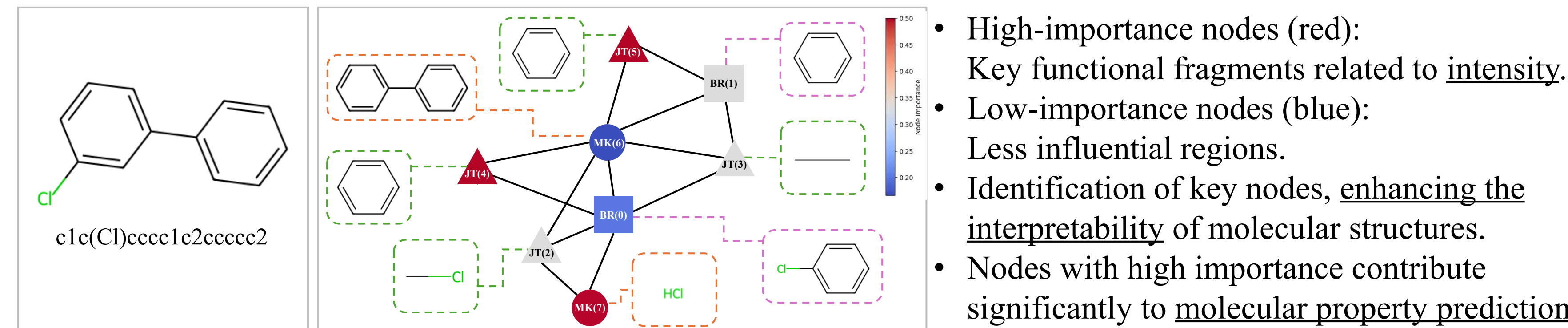
4. Ablation Study

Method	BBBP	HIV	BACE	ESOL	FreeSolv	Lipo
w/o BRICS	0.929±0.009	0.723±0.032	0.570±0.002	0.542±0.034	0.692±0.092	0.839±0.037
w/o JT	0.875±0.012	0.703±0.028	0.543±0.032	1.030±0.079	1.081±0.104	1.016±0.034
w/o Murcko	0.902±0.009	0.698±0.011	0.570±0.026	1.043±0.060	1.079±0.113	1.012±0.032
Ours	0.929±0.014	0.738±0.015	0.597±0.011	0.521±0.014	0.582±0.036	0.785±0.026

2. UMAP Visualization of Embedding Improvement before and after Fine-tuning



3. Node Importance in Triple-View Graphs



Original SMILES

Triple-View Graph

- Removing any of BRICS, JT, or Murcko results in a significant performance drop.
- The full model consistently outperforms ablated versions on both classification and regression tasks.
- Complementary structural information from all three decompositions, making their integration essential.

Conclusion

- Triple-View graph Framework:** Capture chemically meaningful complementary structures via BRICS, JT, and Murcko decomposition, and unification into a single graph topology.
- Interpretability Enhancement:** Attribution enabling intuitive understanding of structural context and identification of the most influential components.
- Limitation:** Dominance of large substructures in the Triple-view graph, overshadowing of small but chemically important regions, and improving domain-specific importance for attribution bias correction.

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